

Dataset Statistics

Table 1: Statistics of the datasets used in our experiments. $|E|$, $|R|$, and $|A|$ denote the numbers of entities, relations, and attributes, respectively. Rel. and Attr. triples denote relation and attribute triples. Gold links denote reference alignments.

Dataset	Subset	KG	$ E $	$ R $	$ A $	Rel. triples	Attr. triples	Gold links
OpenEA [1]	D-W-15K-V1	DBpedia	15,000	248	342	38,265	68,258	15,000
		Wikidata	15,000	169	649	42,746	138,246	
	D-W-15K-V2	DBpedia	15,000	167	175	73,983	66,813	15,000
		Wikidata	15,000	121	457	83,365	175,686	
DBP15K [2]	ZH-EN	ZH	19,388	1,701	8,113	70,414	379,684	15,000
		EN	19,572	1,323	7,173	95,142	567,755	
	JA-EN	JA	19,814	1,299	5,882	77,214	354,619	15,000
		EN	19,780	1,153	6,066	93,484	497,230	
	FR-EN	FR	19,661	903	4,547	105,998	528,665	15,000
		EN	19,993	1,208	6,422	115,722	576,543	
OAEI [3]	Mem-ST	Memory Alpha	254,537	180	287	2,096,198	430,730	1,779
		Star Trek Expanded Universe	55,402	133	194	412,179	155,207	
	Star-SWT	Star Wars Wiki	536,869	561	603	6,675,247	1,570,786	1,429
		The Old Republic Wiki	22,791	137	346	105,543	40,605	
DBP1M [†] [4]	EN-DE	EN	1,122,827	597	1	6,213,639	1,122,827	402,127
		DE	555,981	241	1	1,994,876	555,981	
	EN-FR	EN	1,275,304	603	1	7,031,172	1,275,304	444,715
		FR	647,946	380	1	2,997,457	647,946	
DBpedia-YAGO	–	DBpedia	20,773,519	634	673	30,498,496	36,547,846	5,465,219
		YAGO	5,643,861	63	39	22,306,153	25,290,977	

[†] DBP1M originally contains only relation triples. In our experiments, one entity-name literal triple is added per entity, using a separate label predicate for each KG. The attribute counts above include these added triples.

Table 2: Additional statistics of the OAEI knowledge graph alignment datasets. Instances, classes, and properties are reported separately. Gold alignments are reference correspondences for each category.

Dataset	KG	Graph elements			Gold alignments		
		Instances	Classes	Properties	Instances	Classes	Properties
Mem-ST	Memory Alpha	45,828	181	325	1,358	15	56
	Star Trek Expanded Universe	13,426	283	202			
Star-SWT	Star Wars Wiki	145,033	269	700	1,725	13	41
	The Old Republic Wiki	4,180	101	368			

Table 1 and Table 2 summarize the statistics of the datasets used in our experiments. Following AttrGNN [5], we use the DBP15K version enriched with attribute values from the DBpedia dump (2016-10). Since DBP1M originally contains no literal attributes, we add one entity-name literal per entity, using a separate label predicate for each KG, to enable FLORA+ to run and ensure comparability with methods using entity name information. We convert the resulting graphs to Turtle (.ttl) format for direct loading by FLORA+. We also construct a DBpedia-YAGO dataset from DBpedia (2022-12-01) [6] and YAGO 4.5 [7], with reference alignments derived from shared Wikidata identifiers. We only remove non-English literals, cross-graph identity links, and media and web metadata.

References

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